

Name:

Date:

Period:

Egg Drop Lab

Goal:

We will be dropping your egg from the balcony by the library stairwell. Your goal is to design a device that will prevent the egg from breaking.

Materials:

- 1 egg
- 5 sheets of printer paper
- 1 meter of masking tape

Design

1. The egg will break if too much force is applied to it. Therefore, your device should attempt to decrease the amount of force applied to the egg. Consider the equation $\Sigma F = \Delta p / \Delta t$. Which two **physics quantities** affect the amount of force that the egg feels? How does each quantity affect it?
2. Design a device that you think will keep the egg from breaking. Sketch your device.
3. Explain how your device addresses what you discussed in #1.

Results

4. Explain what happened. Were you successful?

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Lab Report

On Schoology, you will turn in a Lab Report on a Word document that addresses the following:

1. **Purpose**
2. **Your design.** Explain your design and how you expected it to work. Make sure to explain how your device intended to address momentum and/or impulse.
3. **Results.** After you drop the egg, explain what happened. Were you successful?
4. **Conclusion.** Explain how the results differed from what you expected and what you could do differently to improve your design. Make sure to draw a connection between your suggested improvements and the concepts of impulse or momentum.

Please note, every member of your group must turn in a lab report. This is an *individual* assignment.

Grading Rubric

Application to Momentum and Impulse Your design discussion and your conclusion discussion accurately and completely draws the connection between the lab and impulse/momentum concepts.

- 10/10 Meets Standard
- 7/10 You address impulse/momentum, but your discussion is not quite right
- 5/10 Impulse/Momentum is superficially addressed
- 0/10 You do not address momentum or impulse

Completeness Your design and conclusion sections show that you have put some thought into this

- 5/5 Meets Standard
- 3/5 Needs work
- 0/5 These sections are missing or extremely sparse on details

Bonus Your device was successful.

- +3 Your egg survived
- +0 Your egg broke

Total Score _____ / 15